

Commercial Insurance Policy Comparison & Risk Insights Platform

1. Executive Summary

Large commercial insurance brokers manage multiple policy quotations from different insurers while advising enterprise clients on coverage selection. Each insurer structures policy wording, exclusions, endorsements, deductibles, and coverage limits differently, making manual comparison time-consuming and prone to oversight.

The proposed platform digitizes the policy comparison process by extracting policy clauses, standardizing terminology, identifying coverage gaps, and generating structured comparison reports. It enables brokers to make faster, more consistent, and auditable recommendations while reducing operational effort and Errors & Omissions (E&O) risk.

2. Business Context

Commercial insurance brokers routinely evaluate multiple policy documents before recommending the most suitable coverage for corporate clients. The process involves reviewing hundreds of clauses across multiple insurers, each using different legal terminology and document structures.

Manual policy analysis creates several operational challenges:

- Time-intensive comparison of lengthy policy documents.
- Inconsistent interpretation of clauses across brokers.
- Difficulty identifying hidden exclusions and coverage gaps.
- Limited audit trail supporting recommendation decisions.
- Increased regulatory and professional liability exposure.

The objective is to create a centralized platform that assists brokers with policy comparison while preserving human decision-making for final recommendations.

3. Business Problem Statement

The existing policy comparison process relies heavily on manual document review and spreadsheet-based analysis.

As policy complexity increases, brokers spend significant time normalizing terminology, identifying exclusions, validating endorsements, and documenting recommendations.

This results in:

- Longer client turnaround times.
- Higher operational costs.
- Increased possibility of recommendation inconsistencies.
- Limited traceability of advisory decisions.
- Greater Errors & Omissions (E&O) exposure.

4. Project Objectives

The platform aims to:

- Reduce manual effort involved in policy comparison.
- Standardize policy clauses across multiple insurers.
- Improve identification of coverage gaps and exclusions.
- Provide transparent audit trails for advisory decisions.
- Enable faster, evidence-based client recommendations.
- Improve operational efficiency while supporting regulatory compliance.

5. Success Metrics (KPIs)

KPI	Current	Target
Average Policy Comparison Time	4–6 hours	< 1 hour
Manual Review Effort	High	Reduced by 60%
Coverage Gap Detection Accuracy	Manual	Standardized
Policy Recommendation Turnaround	2–3 days	Same Day
Audit Traceability	Partial	100%
Manual Overrides with Justification	Not Tracked	100% Logged

6. Stakeholders

Stakeholder	Responsibility
Insurance Broker	Compare policies and recommend coverage
Enterprise Client	Review recommendations and select policy
Product Manager	Prioritize business requirements
Business Analyst	Gather requirements and define functional specifications
Underwriting Team	Validate policy terms and coverage
Compliance Team	Ensure regulatory and governance adherence
Engineering Team	Develop platform features and integrations
QA Team	Validate business requirements through testing

7. Project Scope

In Scope

- Multi-insurer policy upload.
- Policy clause extraction.
- Coverage comparison.
- Exclusion analysis.
- Recommendation workflow.
- Audit trail.
- Reporting dashboards.

Out of Scope

- Policy issuance.
- Premium payment processing.
- Claims management.
- Underwriting decisions.
- Regulatory filing.

8. Assumptions

- Policy documents are available in supported digital formats.
- Standard clause taxonomy is maintained centrally.
- Brokers remain responsible for final advisory decisions.
- External integrations are available where required.

9. Dependencies

- Policy repository.
- Clause extraction service.
- Authentication platform.
- Reporting database.
- Notification service.

10. Current Process (AS-IS)

Existing Workflow

Client submits multiple insurance policies

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Broker manually reviews policy documents

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Coverage clauses extracted manually

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Comparison performed using Excel

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Coverage gaps identified manually

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Broker prepares recommendation

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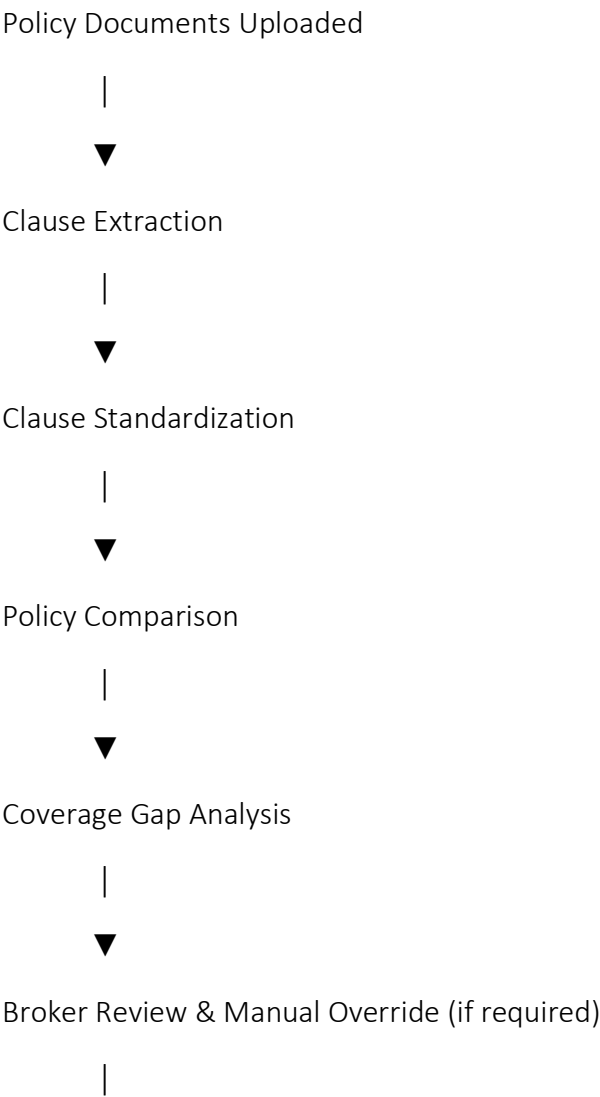
Recommendation shared with client

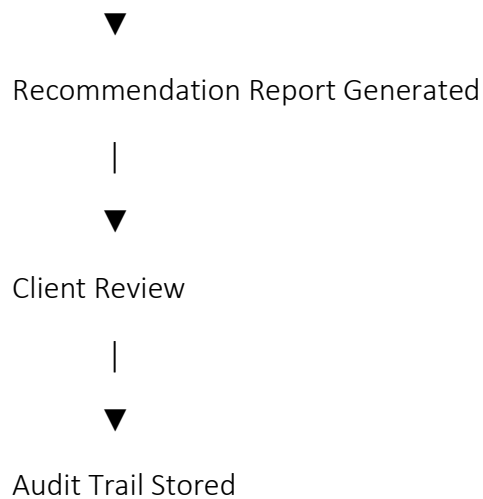
Current Challenges

Challenge	Business Impact
Manual policy comparison	High turnaround time
Different insurer terminology	Inconsistent interpretation
Spreadsheet-based tracking	Version control issues
No standardized clause mapping	Missed coverage differences
Limited audit trail	Higher E&O exposure
Manual recommendation process	Reduced productivity

11. Future Process (TO-BE)

Proposed Workflow





12. Business Process Improvements

Current State	Future State
Manual comparison	Automated comparison support
Different clause terminology	Standardized clause taxonomy
Spreadsheet tracking	Centralized comparison platform
Manual gap identification	Automated gap highlighting
Limited traceability	Complete audit history
Manual reporting	Structured recommendation reports

13. Stakeholder Interaction Matrix

Stakeholder	Inputs	Outputs
Enterprise Client	Policy requirements	Final recommendation
Insurance Broker	Policy documents	Advisory report
Underwriter	Policy terms	Coverage validation
Compliance Team	Regulatory rules	Compliance approval
Business Analyst	Business requirements	Functional specifications
Product Manager	Product roadmap	Prioritized backlog
Engineering Team	Technical implementation	Platform features
QA Team	Test scenarios	Release validation

14. Primary Personas

Persona 1 – Insurance Broker

Goals

- Compare policies efficiently
- Recommend optimal coverage
- Reduce manual effort

Pain Points

- Large policy documents
- Different insurer terminology
- Tight client timelines

Persona 2 – Enterprise Client

Goals

- Understand policy differences
- Select appropriate coverage
- Reduce business risk

Pain Points

- Complex legal wording
- Difficult comparison
- Limited visibility into exclusions

Persona 3 – Compliance Manager

Goals

- Ensure advisory compliance
- Maintain audit records
- Reduce regulatory exposure

Pain Points

- Missing audit history
- Manual oversight
- Inconsistent documentation

15. Expected Business Outcomes

Objective	Expected Outcome
Faster comparison	Reduced broker effort
Standardized clauses	Consistent recommendations
Better traceability	Improved audit readiness
Coverage visibility	Better client decisions
Centralized workflow	Improved operational efficiency

16. Functional Requirements (FR)

Module 1 – Policy Upload & Document Management

FR-01: The system shall allow brokers to upload one or more commercial insurance policy documents in PDF or DOCX format.

FR-02: The system shall validate supported file types and reject unsupported or corrupted documents.

FR-03: The system shall assign a unique Policy ID to every uploaded document.

Module 2 – Clause Extraction & Standardization

FR-04: The system shall extract coverage clauses, exclusions, deductibles, endorsements, and limits from uploaded policy documents.

FR-05: The system shall map insurer-specific terminology to a standardized clause taxonomy for consistent comparison.

FR-06: The system shall retain the original clause reference to support audit traceability.

Module 3 – Policy Comparison

FR-07: The system shall compare two or more uploaded policies side by side.

FR-08: The system shall highlight differences in coverage, exclusions, deductibles, and limits.

FR-09: The system shall identify missing or additional coverages across policies.

Module 4 – Recommendation Workflow

FR-10: The system shall generate a structured comparison report summarizing key differences and identified coverage gaps.

FR-11: Brokers shall be able to review recommendations before sharing them with clients.

FR-12: Manual changes to recommendations shall require a justification and be recorded in the audit trail.

Module 5 – Reporting & Audit

FR-13: The system shall maintain an audit trail for all policy uploads, comparisons, manual overrides, and recommendations.

FR-14: Authorized users shall be able to filter reports by client, insurer, policy type, and comparison date.

FR-15: The system shall provide operational dashboards for tracking comparison requests and processing status.

17. Non-Functional Requirements (NFR)

Category	Requirement
Performance	Standard policy comparison should complete within 60 seconds for supported document sizes.
Availability	System availability target of 99.5% during business hours.
Security	Role-Based Access Control (RBAC) for brokers, compliance, and administrators.
Auditability	Every recommendation and manual override shall be timestamped and attributable to a user.
Scalability	Support concurrent comparison requests without impacting response times.
Usability	Interface should allow users to compare policies without requiring technical expertise.
Reliability	Failed uploads shall be logged with descriptive error messages.
Compliance	Sensitive client information shall be handled in accordance with applicable data privacy requirements.

18. Business Rules

BR-01 – Policy Comparison Eligibility

Only policies belonging to the same insurance product category (e.g., Property, Cyber, Professional Liability) can be compared.

BR-02 – Recommendation Approval

Only users with the Broker role may approve and share final recommendations with clients.

BR-03 – Manual Override

Any manual modification to an automatically generated recommendation must include a mandatory justification.

BR-04 – Audit Trail

Every significant action—including uploads, comparisons, overrides, and report generation—must be recorded with user ID and timestamp.

BR-05 – Coverage Gap Flagging

Coverage differences classified as Critical must be highlighted in the comparison report.

BR-06 – Document Integrity

Uploaded policy documents cannot be modified after successful processing. Any revised document must be uploaded as a new version.

BR-07 – Recommendation Versioning

Each recommendation shall maintain version history to preserve previous advisory decisions.

BR-08 – Access Control

Only authorized users may view client-specific policy documents and recommendation reports.

19. Assumptions & Constraints

Assumptions

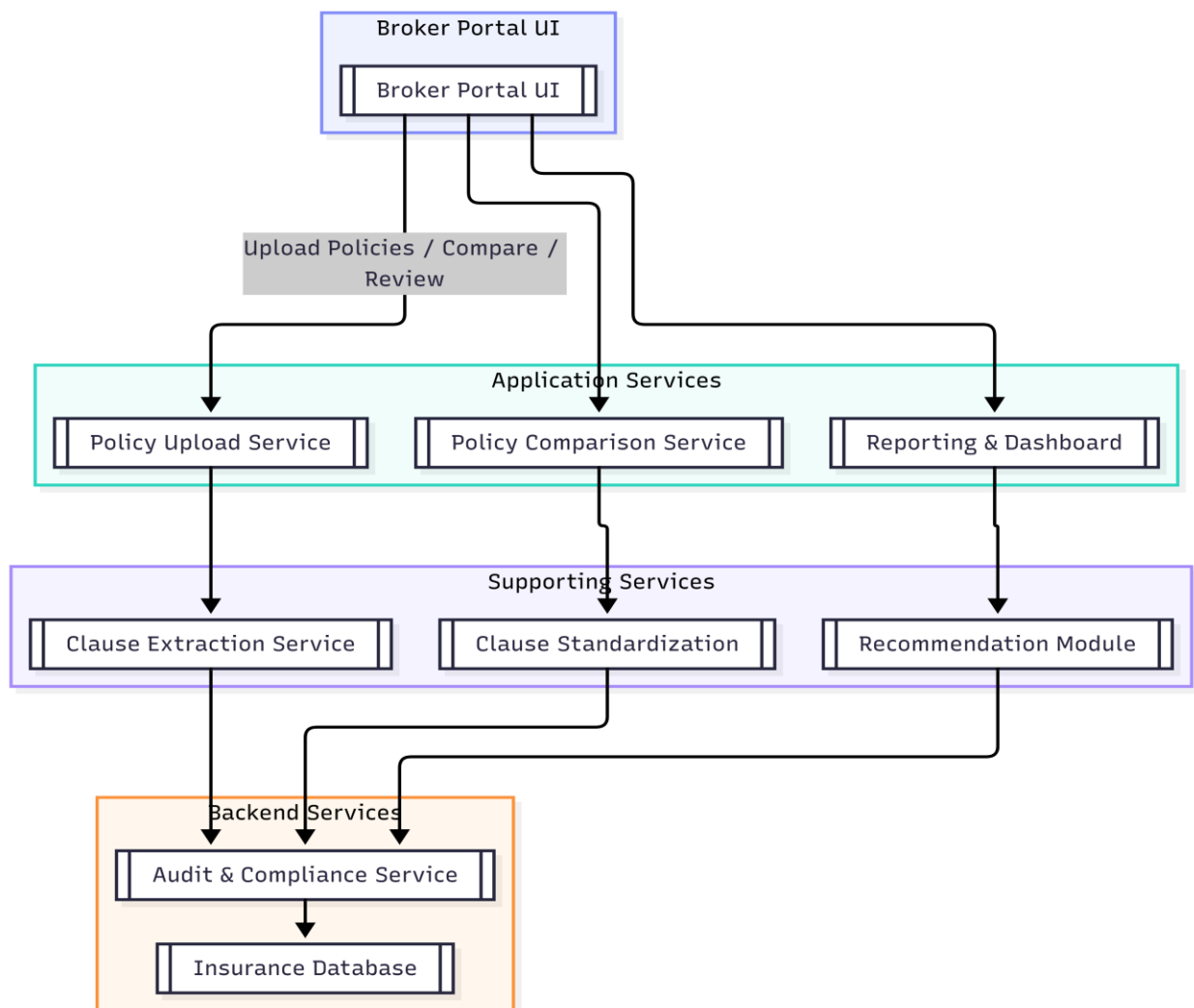
- Policy documents are available in digital format.
- Standardized clause taxonomy is maintained.
- Users receive appropriate role-based permissions.

Constraints

- Comparison is limited to supported document formats.
- Final recommendations remain subject to broker approval.
- External document extraction services must be available for automated processing.

20. High-Level Solution Architecture

Commercial Insurance Policy Comparison Platform



21. Entity Relationship Diagram (ERD)

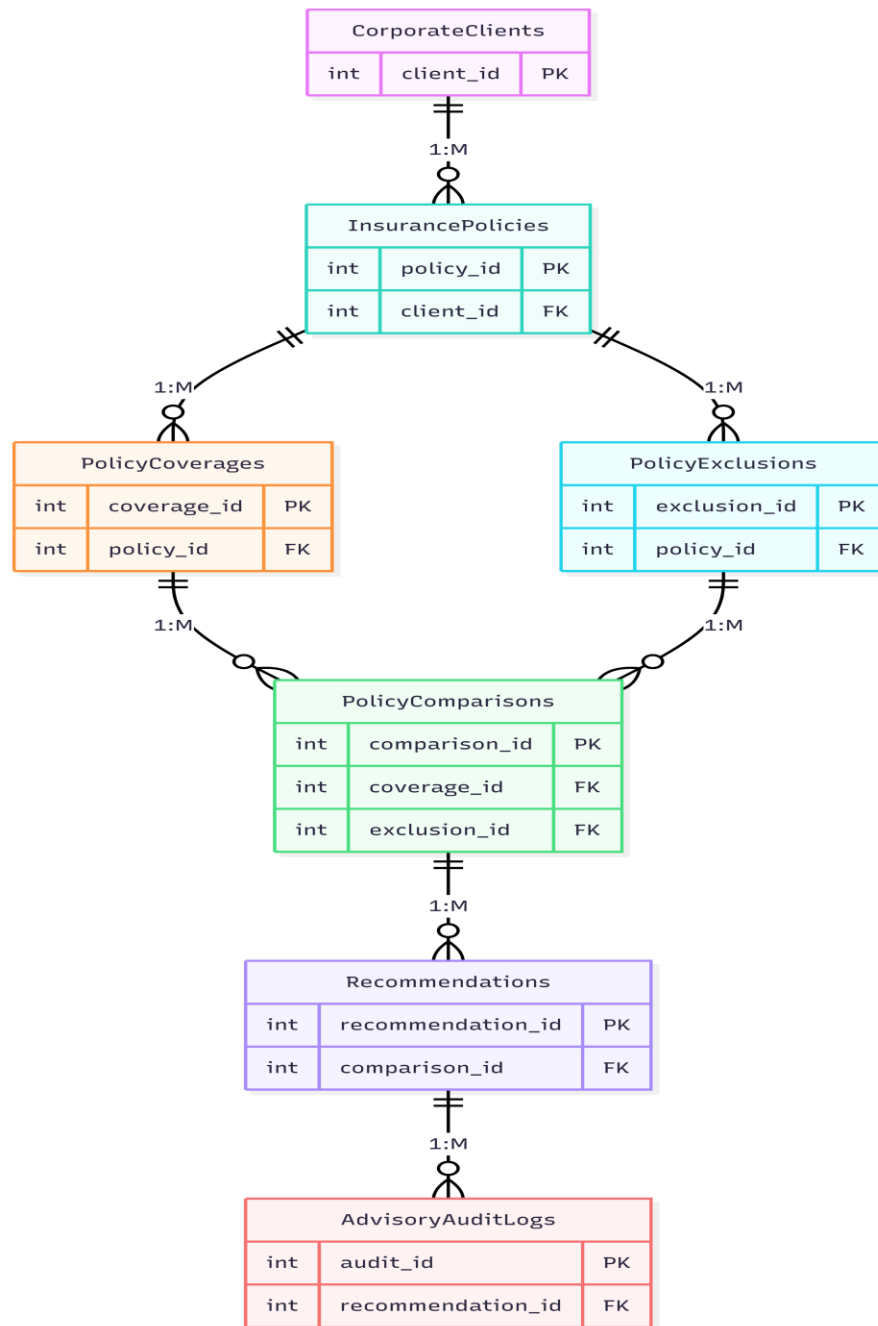


Table Relationships

Parent	Child	Relationship
Corporate Clients	Insurance Policies	One-to-Many
Insurance Policies	Policy Coverages	One-to-Many
Insurance Policies	Policy Exclusions	One-to-Many

Insurance Policies	Policy Comparisons	One-to-Many
Policy Comparisons	Recommendations	One-to-One
Recommendations	Advisory Audit Logs	One-to-Many

22. Data Dictionary

Corporate Clients

Field	Type	Description
client_id	INT (PK)	Unique client identifier
company_name	VARCHAR	Client organization
industry_sector	VARCHAR	Industry classification
risk_profile	VARCHAR	Risk tier

Insurance Policies

Field	Type	Description
policy_id	INT (PK)	Unique policy
client_id	INT (FK)	Client reference
insurer_name	VARCHAR	Insurance carrier
policy_type	VARCHAR	Product category
premium	DECIMAL	Premium amount
upload_date	DATE	Upload timestamp

Policy Coverages

Field	Type	Description
coverage_id	INT (PK)	Coverage identifier
policy_id	INT (FK)	Policy reference
coverage_name	VARCHAR	Coverage description
coverage_limit	DECIMAL	Coverage amount

Policy Exclusions

Field	Type	Description
exclusion_id	INT (PK)	Exclusion identifier

policy_id	INT (FK)	Policy reference
exclusion_name	VARCHAR	Exclusion clause
severity	VARCHAR	Critical / Medium / Low

Policy Comparisons

Field	Type	Description
comparison_id	INT (PK)	Comparison identifier
base_policy_id	INT (FK)	Primary policy
comparison_policy_id	INT (FK)	Compared policy
comparison_date	DATE	Execution date
comparison_status	VARCHAR	Completed / Pending

Recommendations

Field	Type	Description
recommendation_id	INT (PK)	Recommendation ID
comparison_id	INT (FK)	Comparison reference
recommendation_type	VARCHAR	Best Match / Gap Identified
recommendation_status	VARCHAR	Draft / Approved

Advisory Audit Logs

Field	Type	Description
audit_id	INT (PK)	Audit identifier
recommendation_id	INT (FK)	Recommendation reference
action	VARCHAR	Created / Updated / Approved
performed_by	VARCHAR	User
action_timestamp	TIMESTAMP	Audit timestamp

23. Requirement Traceability Matrix

Business Requirement	Functional Requirement	User Story	UAT
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Upload policies	FR-01	US-01	TC-01
Extract clauses	FR-04	US-03	TC-02
Compare policies	FR-07	US-05	TC-03
Detect coverage gaps	FR-09	US-06	TC-04
Generate recommendation	FR-10	US-08	TC-05
Manual override	FR-12	US-09	TC-06
Maintain audit trail	FR-13	US-10	TC-07
Dashboard reporting	FR-15	US-12	TC-08

24. Product Backlog Structure & User Stories

EPIC 1 – Policy Upload & Document Management

US-01 – Upload Insurance Policies

As an Insurance Broker

I **want to** upload one or more insurance policy documents

So that I can compare multiple policy options for my client.

Acceptance Criteria (Gherkin)

Given valid policy documents are available

When the broker uploads the files

Then the system stores the documents

And assigns a unique Policy ID.

US-02 – Validate Policy Documents

As a Broker

I **want** the system to validate uploaded files

So that unsupported or corrupted documents are rejected.

EPIC 2 – Policy Analysis & Comparison

US-03 – Extract Policy Clauses

As a Broker

I **want** the system to extract policy clauses
So that manual review effort is reduced.

US-04 – Standardize Policy Clauses

As a Business User

I **want** insurer-specific terminology mapped to common business terms
So that policy comparisons remain consistent.

US-05 – Compare Insurance Policies

As a Broker

I **want** policies displayed side-by-side
So that differences are easily identified.

US-06 – Identify Coverage Gaps

As a Broker

I **want** missing or reduced coverage highlighted
So that I can recommend the most appropriate policy.

EPIC 3 – Recommendation & Advisory

US-07 – Generate Recommendation Report

As a Broker

I **want** the system to generate a structured comparison report
So that I can discuss recommendations with the client.

US-08 – Review Recommendations

As a Broker

I **want** to review recommendations before sharing them
So that I remain responsible for the final advisory decision.

US-09 – Manual Override with Justification

As a Broker

I **want** to manually modify recommendations

So that business judgment can be applied where necessary.

Acceptance Criteria (Gherkin)

Given a recommendation has been generated

When the broker edits the recommendation

Then the system requires a justification

And records the change in the audit log.

EPIC 4 – Governance & Compliance

US-10 – Maintain Audit Trail

As a Compliance Manager

I want every recommendation change recorded

So that advisory decisions remain fully traceable.

US-11 – Notification & Status Updates

As a Broker

I want notifications when comparisons are complete

So that I can continue my workflow without repeatedly checking the system.

EPIC 5 – Analytics & Reporting

US-12 – Executive Dashboard

As a Business Manager

I want operational KPIs displayed in a dashboard

So that I can monitor workload, turnaround time, and comparison quality.

25. Definition of Ready (DoR)

A user story is considered ready when:

- Business objective is defined.
- Stakeholder is identified.
- Acceptance criteria are documented.
- Dependencies are known.

- Business value is understood.

26. Definition of Done (DoD)

A user story is considered complete when:

- Functional requirements are implemented.
- Acceptance criteria are satisfied.
- UAT is passed.
- Audit logging is validated.
- Product Owner signs off.

27. API Contract – Policy Upload API

Purpose

Allows brokers to upload commercial insurance policy documents for processing.

Method: POST

Endpoint:

/api/v1/policies/upload

Sample Request

```
{  
  "clientId": 101,  
  "policyType": "Cyber Liability",  
  "insurer": "ABC Insurance",  
  "documentName": "CyberPolicy.pdf"  
}
```

Sample Response

```
{  
  "policyId": 501,  
  "status": "Uploaded",  
  "message": "Policy uploaded successfully."  
}
```

28. API Contract – Policy Comparison API

Purpose

Compares two uploaded insurance policies and highlights differences.

Method: POST

/api/v1/policies/compare

Sample Request

```
{  
  "basePolicyId": 501,  
  "comparisonPolicyId": 502  
}
```

Sample Response

```
{  
  "comparisonId": 301,  
  "status": "Completed",  
  "coverageGaps": 4,  
  "criticalExclusions": 2  
}
```

29. API Contract – Notification API

Purpose

Sends notification once policy comparison is completed.

Method: POST

/api/v1/notifications/send

Sample Request

```
{  
  "brokerId": 205,
```

```
"comparisonId": 301,  
"status": "Completed"  
}
```

Sample Response

```
{  
  "notificationStatus": "Sent"  
}
```

30. User Acceptance Testing (UAT)

Test ID	Scenario	Expected Result
TC-01	Upload valid policy	Policy uploaded successfully
TC-02	Upload invalid file	Validation error displayed
TC-03	Compare two policies	Comparison report generated
TC-04	Detect coverage gaps	Missing coverages highlighted
TC-05	Generate recommendation	Recommendation created
TC-06	Manual override	Justification captured in audit log
TC-07	Dashboard refresh	Latest KPIs displayed
TC-08	Notification	Broker receives completion notification

31. UAT Sign-Off

Role	Name	Status
Business Owner	To be Assigned	Pending
Product Manager	To be Assigned	Pending
Business Analyst	Vaibhav Yadav	Completed
QA Lead	To be Assigned	Pending

32. Governance & Compliance

Audit & Traceability

- Every recommendation is linked to its originating policy comparison.
- Manual overrides require mandatory justification.

- User actions are recorded with timestamp and user ID.

Role-Based Access Control

Role	Permissions
Broker	Upload, compare, recommend
Compliance Manager	Review audit logs
Product Manager	View reports and KPIs
Administrator	Manage users and configurations

Data Governance

- Version control maintained for policy documents.
- Recommendation history preserved.
- Audit logs retained for compliance reviews.
- Sensitive client information accessible only to authorized users.

33. Project Risk Register

Risk	Impact	Mitigation
Poor document quality	Incorrect clause extraction	Validate uploaded documents before processing
Inconsistent policy wording	Comparison inaccuracies	Standardized clause taxonomy
Manual override misuse	Incorrect recommendations	Mandatory justification and audit trail
Delayed comparisons	SLA impact	Dashboard monitoring and workload tracking
Unauthorized access	Compliance risk	Role-Based Access Control (RBAC)

34. SQL Analytics Overview

The database supports operational reporting and decision-making by enabling analytics across policy comparison, coverage analysis, broker productivity, and compliance monitoring.

The SQL scripts included in the repository demonstrate how business requirements can be translated into actionable insights for stakeholders.

Key Analytics Areas

- Policy processing performance
- Coverage gap analysis
- Broker productivity
- Manual override monitoring
- Audit & compliance reporting
- Client portfolio insights

35. Sample Business Analytics Queries

Query	Business Purpose
Total Policies Processed	Measure operational workload
Policy Comparison Turnaround Time	Monitor SLA performance
Coverage Gap Analysis	Identify frequently missing coverages
Manual Override Report	Track broker interventions
Critical Exclusion Report	Monitor high-risk policies
Client Risk Distribution	Segment clients by risk profile
Broker Productivity	Measure comparison volume by broker
Audit Activity Summary	Review governance and compliance

Repository Files

- insurance_schema.sql
- insurance_sample_data.sql
- insurance_analytics_queries.sql
- commercial_insurance_dataset.xlsx
- tableau_dashboard.png

36. Dashboard Design

Executive Dashboard

KPI Cards

- Average Policy Comparison Turnaround Time

Operational Visuals

- SLA Scorecard

- Exclusion Density
- Audit Ledger
- Policy Comparison Trend

Dashboard Filters

- Underwriter Carrier
- Policy Type
- Unmitigated Risk Severity
- Year of Document Upload

37. Business KPIs

KPI	Business Value
Average Comparison Time	Operational efficiency
Coverage Gap Detection Rate	Recommendation quality
Manual Override Percentage	Process consistency
Audit Compliance Rate	Governance effectiveness
Broker Productivity	Team performance
Recommendation Turnaround Time	Client responsiveness

38. Expected Business Benefits

Operational Benefits

- Reduced manual policy review effort.
- Faster comparison of multiple insurer policies.
- Standardized policy evaluation process.
- Improved broker productivity.

Business Benefits

- More consistent advisory recommendations.
- Better visibility into coverage gaps.
- Improved client confidence through structured comparisons.
- Enhanced decision support for brokers.

Governance Benefits

- Complete audit trail for recommendation changes.
- Improved regulatory readiness.
- Better traceability of advisory decisions.
- Reduced Errors & Omissions (E&O) exposure.

39. Future Roadmap

Phase 1

- Policy upload
- Clause extraction
- Policy comparison

Phase 2

- Recommendation workflow
- Executive dashboards
- Audit reporting

Phase 3

- AI-assisted recommendation suggestions
- Predictive risk insights
- Carrier benchmarking
- Client portfolio analytics

40. Project Deliverables

Documentation

- Business Requirements Document (BRD)
- Functional Requirements
- Non-Functional Requirements
- Business Rules
- User Stories
- Gherkin Acceptance Criteria

- API Contracts
- UAT
- Governance Documentation

Technical Assets

- ERD
- Data Dictionary
- SQL Schema
- Sample Dataset
- Analytics Queries
- Tableau Dashboard
- GitHub Repository

41. Project Summary

Business Problem

Commercial insurance brokers spend significant time manually comparing policy documents, resulting in inconsistent recommendations, limited traceability, and increased operational effort.

Proposed Solution

A centralized Policy Comparison & Risk Insights Platform that standardizes policy clauses, highlights coverage gaps, supports broker decision-making, and maintains a complete audit trail for governance and compliance.

Value Delivered

- Reduced manual effort
- Faster policy comparison
- Improved recommendation quality
- Better governance and auditability
- Enhanced client decision support